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GGG-H-142c

December 18, 1961

SUPERSEDING
Fed. Spec. GGG-H-142b
October 6, 1953

FEDERAL SPECIFICATION

HELMET, CONSTRUCTION WORKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers safety helmet for protection of the wearer's head against impacts of falling or flying objects. The helmets are intended for protection against these hazards in mining, building construction, shipbuilding, stevedoring, tunneling, timber work, and manufacturing.

1.2 Classification

1.2.1 *Types.* The helmets shall be furnished in the following types, as specified:

Type I.—Full brim.
Type II.—Brimless with visor.

1.2.2 *Colors.* When specified, the helmets shall be furnished in the colors as specified in 3.5, 3.5.1 and 6.1.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-406—Plastics, Organic; General Specifications, Test Methods.
UU-P-268—Paper, Kraft, Untreated Wrapping.
CCC-A-700—Artificial Leather, Cloth, Coated Vinyl Resin (Upholstery).
CCC-F-451—Flannel; Canton.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-T-76—Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).
PPP-T-97—Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-C-5039—Cloth, Cotton Twill.
MIL-B-10377—Box, Wood Cleated, Veneer, Paper Overlaid.
MIL-L-10547—Liners, Case, Waterproof.

FSC 8415

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The safety helmets furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Component parts. Each helmet shall consist of a shell, headband, crown straps forming a cradle, hammock or a crown suspension for supporting the helmet on the wearer's head, and if specified in the contract or order an adjustable chinstrap and a winter lining (see 6.1).

3.3 Material. All materials coming into contact with the skin shall be nonirritating.

3.3.1 *Shell.* The shell shall be of smooth, hard material which shall conform to the requirements specified herein. Materials used in the fabrication of the helmet shells furnished under this specification shall be at the option of the manufacturer, provided helmets of this material have been tested and passed the qualification tests specified herein. When specific materials are required by the ordering activity (see 6.1), the materials requested shall be provided only in helmets of this material which have been tested and passed the qualification test requirements herein specified.

3.3.2 *Headband.* Unless otherwise specified, the headband shall be genuine vegetable-tanned leather, full grained and soft. The headband shall be smoothly finished on the surface which will be in contact with the head. If specified in the contract or order (see 6.1), headband may be made of artificial leather conforming to class 3 of CCC–A–700.

3.3.3 *Crown straps.* Crown suspension or straps shall be of webbing, closely woven, or other suitable materials.

3.3.4 *Accessory materials.* If chinstraps and winter linings are specified in the contract or order, they shall be made of the following materials:

3.3.4.1 *Chinstrap.* The chinstrap shall be adjustable and shall be of closely woven cotton or linen webbing or of genuine leather.

3.3.4.2 *Winter lining.* The winter lining shall consist of two-ply (layers) material. The outer layer shall conform to the requirements of MIL–C–5039 and the inner layer to the requirements of type A of CCC–F–451.

3.4 Construction.

3.4.1 *Shell.* The shell shall be of one-piece seamless construction, dome shaped to provide the inside clearances specified in 3.4.2.2. Type I helmet shells shall have a full brim, while type II helmet shells shall be brimless with a visor.

3.4.1.1 *Metal parts.* Metal parts shall not impair the electrical protection which the shell provides the wearer. They shall be so placed that no part of the worker's head is likely to be scratched by a blow on the brim of the helmet which would cause the helmet to slide down on the ears or forehead. Metal parts shall be rustproofed.

3.4.1.2 *Brim.*

3.4.1.2.1 *Type I, full brim.* The brim for type I helmets shall be an integral extension of the shell dome, and shall be continuous around the helmets. The width of the brim shall be not less than 1½ inches and not more than 3 inches measured from the inside edge of the shell, except that the width at the front may taper to not less than 1¼ inches. With the helmet held in a horizontal position, the brim shall slope downward in a straight line at an angle of not less than 15 degrees nor more than 37 degrees from the horizontal.

3.4.1.3 *Visor.*

3.4.1.3.1 *Type II, brimless with visor.* The visor for type II helmets shall extend forward from the crown not less than 1½ inches, and

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shall be not less than 5½ inches wide at the peak.

3.4.2 *Headbands.* Headbands shall be adjustable and shall permit adjustment to all sizes within the range marked on helmet and band. The thickness of leather headbands, determined at 10 points uniformly spaced throughout the circumference of the headband, shall be not less than 1/32 inch, when measured with a gage having a 5/16-inch-diameter contact foot under a 1-pound load. The surface of the headband in contact with the head shall be not less than 1¼ inches in width. The headband shall be securely attached either to the helmet shell or to the crown straps for assuring its proper position in the helmet. The manner of the attachment shall not interfere with the adjustments of the crown straps. Padding or stiffener strips, if used as a regular part of the headband, shall be secured to the leather.

3.4.2.1 *Adjustment range.* Headbands shall be adjustable in ⅛-size increments from at least size 6¾ to 7½ inclusive. The sizes shall be such that the circumference of the inside of the headband for size 7 is 22 inches, with a ⅜-inch increment for each ⅛ size above and below size 7. Tolerance of circumferential measurements shall be ±⅛ inch.

3.4.2.2 *Headband clearance.* As determined by the method described in 4.6.8, the clearance between the inside of the shell and the outside of the headband shall be not less than ¼ inch nor more than ¾ inch.

3.4.3 *Crown straps.* Crown straps shall be not less than ½ inch in width. These straps, when adjusted and properly interlaced, shall form a cradle or hammock for supporting the helmet on the wearer's head and holding the helmet away from the head, thus acting as a shock absorber when the helmet is struck. The number of straps shall be adequate and their location in the helmet shall be suitable for the above purpose. A cradle, hammock, or crown suspension shall be formed in one or a combination of the following ways:

(*1*) The upper extremities of the straps formed into loops and securely attached to an adjusting means.

(*2*) Continuous straps across the crown.

(*3*) A molded plastic suspension piece. When worn, the fixed crown-strap assembly shall provide for a minimum clearance of 1¼ inches and no more than 1⅝ inches (maximum). Adjustable crown-strap assemblies shall provide for adjustment of clearance up to at least 1½ inches.

3.4.4 *Accessories.* If chinstraps and winter linings are specified in the contract or order (see 6.1), they shall conform to the following requirements:

3.4.4.1 *Chinstraps.* The adjustable chinstrap shall be of not more than two pieces of webbing or leather of not less than ½ inch in width and not less than 16 inches in adjusted length excluding the attachments to the helmet. The means for adjusting the chinstrap shall assure a secure hold of the helmet on the wearer's head and quick removal of the helmet by releasing the strap. The chinstrap shall also be adjustable for wearing at the back of the head. The webbing shall have no frayed or loose edges that may unravel. The leather, when used in the chinstrap, shall be full grained and smoothly finished on the surface that will be in contact with the chin. All metal parts, including rivets, shall be smooth and free from sharp or rough edges or projections. Rivet heads shall be round.

3.4.4.2 *Winter lining.* The winter lining shall be designed to cover the skull, neck, and ears. The skull cap, part of the winter lining, shall be attached to the helmet cradle or to the helmet in such a manner that it can be readily removed and reattached. The earlug and neck part may be made either in one piece with the skull cap or attachable to it by tabs or a zipper. The neck and earlug shall be drawn close to the face by a lace or a chinstrap.

3.5 **Colors.** When specified by the procuring agency (see 6.1), helmets shall be furnished in color. Colors shall conform to Fed. Std. No. 595 for the color number specified. Department

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of Defense purchases shall be restricted to the following standard colors:

Color	Color Number (Fed. Std. No. 595)*
Maroon	10075
Red	11105
Terra Cotta	10233
Orange	12197
Yellow	13538
Green	14260
Blue	15177
Gray	16492
Black	17038
White	17875

*Individual (3x5 inch) color chips are available from General Services Administration, Business Service Center, Washington 25, D.C.

3.5.1 *Color application.* The color shall be provided by the use of high-quality pigment, pigmented lacquers, alkyd enamel, polyester finish, or other acceptable approved coatings.

3.6 Physical requirements.

3.6.1 *Insulation resistance.* The helmet shall not show a leakage current in excess of 3 milliamperes when tested as specified in 4.6.2.

3.6.2 *Impact resistance.* When tested in accordance with 4.6.3, helmets shall not transmit an average force of more than 850 pounds. Individual specimens shall not transmit a force greater than 1,000 pounds.

3.6.3 *Penetration strength.* The helmet shall be neither dented nor pierced for more than ⅜ inch, nor shall the helmet be forced down over the standard head form, nor shall the straps pull out or break when tested as specified in 4.6.4.

3.6.4 *Edge strength, type II.* When tested in accordance with 4.6.5, type II helmets shall show a deflection under a 40-pound load of not more than ¾ inch and an ultimate strength of not less than 50 pounds.

3.6.5 *Flammability.* The thinnest section of the shell shall not burn at a rate greater than 3 inches per minute when tested as specified in 4.6.6.

3.6.6 *Water absorption.* The helmet-shell material shall absorb not more than 5 percent of water when tested as specified in 4.6.7.

3.6.7 *Weight.* The weight of each complete helmet without winter lining and chinstrap shall not exceed 14½ ounces (see 4.6.9).

3.6.8 *Ventilation.* Provision shall be made between the shell and the headband, or in the headband proper, for a free passage of air for ventilation purposes.

3.7 Interchangeability. The headband and crown-strap assembly shall be easily removable and replaceable, and shall be interchangeable among shells of the same size furnished by the same manufacturer provided that a single type of lining is furnished. Where two types of linings are involved, neither lining need be usable as a replacement in shells originally provided with the other type.

3.8 Instruction sheet. An instruction sheet which explains the proper method of wearing and adjusting the helmet shall be furnished with each helmet. The sheet shall be stowed in the helmet so as to remain where placed until intentionally removed.

3.9 Toxicity. The procuring agency shall determine that the finished item, when used for its intended purpose, is not likely to result in adverse effects on health of personnel. In the case of Army procurement, the Surgeon General of the Army will act as advisor to the procuring agency in making the determinations.

3.10 Serviceability. Helmets procured under this specification shall provide protection, retain color and appearance, and shall be fully serviceable and usable for a minimum period of one year under normal conditions of service.

3.11 Marking.

3.11.1 *Shell.* The inside of each helmet shell shall be marked in a permanently legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.11.2 *Headband.* The range of sizes to which the headband is adjustable shall be marked on labels cemented to, or embossed upon, the helmet shell and the headband (see 3.4.2.1).

3.12 Workmanship. The workmanship shall be first class in every respect. The safety helmets shall have no sharp edges or burrs, and shall be free from any other defects which de-

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tract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification inspection.[1] Qualification inspection shall be conducted at a laboratory satisfactory to the Bureau of Ships. This inspection shall consist of visual and dimensional examination and the tests specified in 4.6, except that the test specified in 4.6.5 applies to type II helmets only.

4.2.1 *Samples for qualification inspection.* Thirty-six in each of the types for which approval is desired shall be furnished.

[1] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

4.3 Sampling for acceptance inspection.

4.3.1 *Lot.* All helmets of one type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.3.2 *Sampling for visual and dimensional examination.* A random sample of helmets shall be selected from each lot in accordance with MIL–STD–105 at inspection level III. The acceptable quality level shall be equal to 2.5 percent defective.

4.3.3 *Sampling for insulation-resistance test (see 4.6.2).* A random sample of helmets shall be selected from each lot in accordance with MIL–STD–105 at inspection level L–7.

4.3.4 *Sampling for impact test (see 4.6.3), penetration test (see 4.6.4), and edge-strength test, type II (see 4.6.5).* From each lot, a random sample of helmets shall be selected separately for each test in accordance with MIL–STD–105 at inspection level L–4.

4.4 Acceptance inspection.

4.4.1 *Visual and dimensional examination.* Each of the sample helmets selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as indicated below. Any helmet in the sample containing one or more visual or dimensional

Examine	Defect	Major	Minor
Finish	Color or type of finish not in accordance with approved sample	X	
Design	Any characteristic not in accordance with the specified requirements or approved sample seriously affecting serviceability.	X	
	Any characteristic not in accordance with the specified requirements or approved sample affecting serviceability but not seriously.		X
Material (General)	Any component not fabricated of approved material seriously affecting serviceability.	X	
	Any component not fabricated of the approved material not seriously affecting serviceability, e.g., rivet is fabricated from steel when brass is specified.		X
Leather, headbands or chinstraps (if applicable).	Not fullgrain	X	
	Side of the leather in contact with skin not smooth		X
	Any deep cuts, holes, open scratches, deep wrinkles or pronounced scars seriously affecting serviceability.	X	
	Any cuts, holes, open scratches, wrinkles, or scars affecting serviceability but not seriously.		X
	Spots or stains that are clearly noticeable		X

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Examine	Defect	Major	Minor
Leather, headbands or chinstraps (if applicable)—Continued	Hard or bony leather or excessively flanky leather seriously affecting serviceability.	X	
	Flanky leather affecting serviceability but not seriously		X
	Any assembly that is not adjustable where required, e.g., chinstrap not adjustable.	X	
	Any sharp burr, sliver, or edge		X
	Any component or assembly malformed seriously affecting serviceability.	X	
	Any component or assembly malformed affecting serviceability but not seriously.		X
	Any fastener, i.e., eyelets, rivets missing or not securely set to a degree where it may become detached from assembly.	X	
	More than one fastener not securely set but is not expected to become detached from assembly.		X
	Any fastener too tightly set or compressed resulting in damage to assembly, e.g., too much pressure applied to rivet resulting in damage to shell.	X	
Assembly highlights	Attachment of the cradle assembly to helmet does not provide adequate clearance between the shell and the headband as compared with approved sample.	X	
Crown straps and laces	Any rip, hole, cut, or tear or otherwise defective seriously affecting serviceability.	X	
	Defective to a degree where serviceability will be affected but not seriously, e.g., misweave, slubs, etc., if applicable.		X
Shell	Any hole, soft spot, air pocket, void, evidence of delamination, rough or abraded area, cracked, chipped, blistered, pitted, or otherwise defective seriously affecting serviceability.	X	
	Scratches, air pocket, void, rough or abraded area, chipped, pitted, blistered or otherwise defective affecting serviceability but not seriously affecting appearance.		X
	Metal parts and metal clips for attaching crown straps and headbands to shell, positioned in manner likely to cause injury if blow on brim forces helmet over ears or forehead.	X	
Construction and workmanship general (applicable to all components, unless otherwise indicated herein).	Any component omitted, damaged, misplaced, operation omitted or not properly performed seriously affecting serviceability.	X	
	Any component misplaced, damaged, operation not properly performed affecting serviceability but not seriously.		X
	Any component not firmly affixed or attached to an extent that parts or assembly may become separated in use.	X	
	Any component not firmly affixed or attached but part or assembly is not expected to become detached in use.		X
Marking	Missing, incomplete, incorrect, illegible, not in specified location or not accomplished in specified manner.		X
Packaging	Packaging not level specified	X	
	Package not as specified		X
Packing	Packing not level specified	X	
	Exterior container not as specified		X
	Closure of exterior container incorrect		X
	Strapping missing where specified or incorrectly applied		X
	Caseliners missing where required or caseliners incorrectly applied		X
	Tape missing or incorrectly applied		X
	Container marking missing, illegible or incorrect	X	

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defects shall be rejected, and if the number of defective helmets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 *Acceptance tests.* Samples selected in accordance with 4.3.3 and 4.3.4 shall be subjected to the tests specified in 4.6.2 through 4.6.5.3. If any sample helmet fails one or more of these tests, the entire lot represented by the sample shall be rejected.

4.5 Rejected lots. Rejected lots may be offered again for inspection provided the contractor has removed all nonconforming helmets. Samples from such resubmitted lots shall again be selected and inspected to verify compliance with this specification.

4.6 Test procedures.

4.6.1 *Preparation of samples.* Where it is evident that the sample helmets under test have a protective coating over the basic material, the entire exterior surface of the shell shall be abraded until the basic material is exposed using number 60-grit garnet paper. All samples shall then be conditioned in an oven for 96 hours at 50°±2° C., then cooled in a desiccator and exposed for 96 hours in an atmosphere having 50±2 percent relative humidity and a temperature of 25°±2° C. Immediately following conditioning, samples shall be subjected to the applicable tests specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5.

4.6.2 *Insulation-resistance test.* The insulation-resistance test shall be performed as follows to determine conformance with 3.6.1:

4.6.2.1 *Apparatus.* The test apparatus shall consist of the following:

(*a*) A vessel containing a solution of 1 ounce of sodium chloride per gallon of water and of sufficient size to submerge an inverted helmet to within ½ inch of the junction of brim with the dome.
(*b*) Wire frame for suspending inverted helmet in sodium-chloride solution.
(*c*) Source of 60-cycle alternating current with 2,200 volts (root mean square).
(*d*) Wiring and terminals for application of voltage across dome of helmet.
(*e*) Voltmeter of sufficient capacity.
(*f*) Milliammeter of sufficient capacity and accuracy to measure the specified current.

4.6.2.2 *Procedure.* An inverted helmet with the head-harness assembly removed shall be placed on the wire frame and submerged in the solution to within ½ inch of the junction of the brim with the dome. The helmet shall also contain the solution to the same level. If the helmet is provided with holes in the crown near the brim, the helmet shall be submerged to within ½ inch of the holes. One terminal from the current source shall be connected to the wire frame which supports the helmet and the other shall be immersed in the sodium-chloride solution which is contained by the helmet. The 2,200 volts shall then be applied for 1 minute. If passage of current is evidenced by deflection of the voltmeter the helmet shall be considered as having failed to conform to 3.6.1. If no deflection of voltmeter occurs, a milliammeter is placed in the circuit and a reading taken.

4.6.2.3 *Report.* For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.6.3 *Impact test.* The impact test shall be performed as follows to determine conformance with 3.6.2:

4.6.3.1 *Apparatus.* The test apparatus shall consist of the following:

(*a*) A standard head form (see 6.4). The standard head forms are of four sizes: Small, medium, large, and extra large. Test-block forms which simulate the standard head forms may be employed. The blocks shall duplicate the crown dimensions and contours from the top down to the ear and eye levels, including the top portion of the ears. Each test block shall have a hole drilled vertically from the geometric center of the crown to facilitate measurement of helmet crown clearance. The geometric center is located by measuring the maximum width at the base of the head form, locating the midpoint and then drawing a straight line through

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this point, bisecting the base from front to back. The midpoint of this line is the geometric center, and a hole is drilled vertically from base to crown. To protect the test block from damage, each head form may be provided with a steel insert. Other accurate means for measuring crown clearance may be used (see 4.6.3.1(*e*)).

(*b*) A Brinell penetrator assembly as shown on figure 1. The impression bar shall be of a metal having a predetermined Brinell hardness of 18 to 30 as measured with a 500-kilogram load using a 10-millimeter ball. The thickness of the bar shall be such that it is at least 10 times the depth of any indentation and the width such that the distance from the center of any indentation to the edge of the bar is at least 2½ times the diameter of the indentation. (An impression bar made of 2 S-O aluminum ½ inch thick and 1⅛ inch wide fulfills these conditions within the specified load range). The Brinell penetrator used in the impact test shall be a hardened-steel ball ½ inch in diameter.

(*c*) A steel ball approximately 3¾ inches in diameter and weighing 7.8 to 8.0 pounds.

(*d*) A Brinell microscope or other suitable micrometer microscope accurate to 0.05 millimeter (mm.).

(*e*) For "fixed" crown type assemblies, the proper head form shall be selected by fitting the sample to the largest possible test block with the headband sufficiently loose to permit a crown clearance of 1$\frac{5}{16}$ inches or less, when a 25-pound weight is placed upon the helmet. If the head form does not permit a clearance of 1$\frac{5}{16}$ inches or less, the next smaller size form shall be employed.

4.6.3.2 *Test conditions.*

4.6.3.2.1 *For qualification tests.* Impact tests shall be conducted with the specimens maintained at temperatures of 20°±2° F. and 120°±2° F. Specimens shall be subjected to the test temperature for at least 4 hours prior to the impact test.

4.6.3.2.2 *For acceptance tests.* Impact tests shall be conducted with the specimens maintained at the prevailing atmospheric temperature.

4.6.3.3 *Size of specimens (for qualification tests).* Specimens shall be tested on all four of the standard head forms.

4.6.3.4 *Procedure.*

(*a*) *Mounting specimens.* The specimens shall be mounted on the appropriate standard head form so that a plane passing through the base of a crown is approximately horizontal. For adjustable crown type assemblies the proper head form shall be selected by fitting a specimen to the largest possible test block and drawing the cradle straps up as tight as possible with the cradle lace untied. The helmet shall fit the head form with the sweatband sufficiently loose to permit a crown clearance of ½ inch or less when a 25-pound weight is placed upon the helmet. If the head form does not permit a clearance of ½ inch or less, the next smaller size form shall be employed. For "fixed" crown-type assemblies, the proper head form shall be selected by fitting the specimen to the largest possible test block with the sweatbands sufficiently loose to permit a crown clearance of 1$\frac{5}{16}$ inches or less when a 25-pound weight is placed upon the helmet. If the head form does not permit a clearance of 1$\frac{5}{16}$ inches or less, the next smaller size form shall be employed.

(*b*) *Crown clearance.* With the proper head form selected in accordance with 4.6.3.4(*a*), the sample shall be adjusted to or have a crown clearance of 1¼+$\frac{1}{16}$−0 inches between the crown of the head form and the underside of the top of the helmet crown when a 25-

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pound weight is placed on the helmet. The clearance may be measured through the drilled hole (see 4.6.3.1(*a*)) with a suitable measuring device or by other accurate methods as applicable. For "fixed" crown type assemblies, the following procedure may be used: Using the apparatus described in 4.6.3.1(*e*), and figure 2, the zero point of the ruler less the thickness of the shell shall be placed even with the top of the weight when the weight is resting on the test block. When a helmet is placed on the test block and the weight lowered on it, the top of the weight will then indicate on the ruler the amount of crown clearance.

(*c*) *Application of impact.* The headform penetrator assembly (see figure 1) shall be placed on a substantially level floor and located beneath the drop ball such that the hole in the center of the head form is aligned with the center of the ball by means of a plumb bob. With the helmet mounted on the head form as specified in 4.6.3.4(*a*) and (*b*), the 8-pound ball shall be dropped vertically on the center of the crown from a height of 5 feet measured from the bottom of the ball to the top of the helmet. Impressions shall be spaced at least 2½ diameters apart, edge to edge. Off-center blows as indicated by elliptical impressions shall be disregarded. The diameter of the impression produced on the impression bar shall then be measured to the nearest 0.1 mm. with the micrometer microscope. For each test, the average Brinell hardness number of the 7-inch impression bar used shall be recorded.

4.6.3.5 *Report.* The average force and the greatest individual force for all helmets tested under one group shall be computed and reported. The forces shall be computed from the impression diameters using the following Brinell formula:

$$F=2.2\times H\times\frac{\pi D}{2}\times(D-\sqrt{D^2-d^2}$$

where:

F is the transmitted force in pounds,
H is the average Brinell hardness number of the impression bar,
D is the diameter of the impression ball in millimeters, and
d is the diameter of the impression in millimeters (see 6.5).

4.6.4 *Penetration test.* The penetration test shall be performed as follows to determine conformance with 3.6.3:

4.6.4.1 *Apparatus.* The test apparatus shall consist of standard head forms or test blocks as specified in 4.6.3.1(*a*) and a 16-ounce plumb bob with a steel point having an included angle of 36 degrees.

4.6.4.2 *Procedure.* The specimen shall be mounted on a test block as specified in 4.6.3.4(*a*) and the plumb bob dropped squarely onto the center of the crown from the height of 10 feet, measured from the bottom point of the plumb bob to the top of the helmet. The depth the plumb bob has pierced the helmet shall be measured to the nearest 1/32 inch and shall include the thickness of the shell material.

4.6.4.3 *Report.* For each specimen, the depth of penetration to the nearest 1/32 inch shall be reported.

4.6.5 *Edge-strength test, type II.* The edge-strength test shall be performed as follows on type II helmets to determine conformance with 3.6.4:

4.6.5.1 *Apparatus.* The test apparatus shall consist of the following:

(*a*) A universal testing machine capable of applying the loading specified in 3.6.4 at a rate of head travel of 0.05±0.01 inch per minute.

(*b*) A dial gage for measuring deflections under loading and accurate to 0.01 inch.

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(*c*) A jig designed to support a type II safety helmet on its side on the bedplate of the testing machine so that a plane intersecting the edge of the crown is vertical and the long axis of the helmet horizontal.

4.6.5.2 *Procedure.* With a specimen mounted in the jig, the traveling head of the test machine shall be brought into contact with the edge of the crown and the rate of deflection adjusted to 0.05±0.01 inch per minute. The dial gage shall be adjusted to measure the travel of the moving head. Deflections shall be measured at intervals of loading of 10 pounds. Loading shall be continued to failure.

4.6.5.3 *Report.* For each specimen, the deflection under a load of 40 pounds and ultimate load at failure shall be reported.

4.6.6 *Flammability test.* The flammability test shall be performed as follows to determine conformance with 3.6.5:

4.6.6.1 *Apparatus.* The test apparatus shall consist of the following:

(*a*) Rigid support and clamps for mounting test strips in specified position.

(*b*) Alcohol lamp or gas burner.

(*c*) Stop watch.

4.6.6.2 *Test conditions.* Test specimens shall be subjected to the flammability test after subjection to the tests specified 4.6.3, 4.6.4, or 4.6.5.

4.6.6.3 *Procedure.* Three strips approximately ½ inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner. These strips shall be cut from a portion of the hat so as to give as flat a section as practicable. Lines shall be drawn across each strip at ½-inch intervals, measured along the surface and starting from one end. The other end shall be clamped to the rigid support so that the longitudinal axis is horizontal and the transverse axis is inclined 45 degrees. The alcohol lamp or gas burner, with a neutral, blue flame ½ to ¾ inch in height, shall be placed under the free end of the strips, so adjusted that the flame tip is just in contact with the material. A stop watch shall be started when the flame is placed in contact with the strip under test. At the end of 30 seconds, the flame shall be removed and the strip allowed to burn noting the rate of burning.

4.6.6.4 *Report.* The rate of burning shall be reported as the average for the three specimens.

4.6.7 *Water-absorption test.* Method 7031 of L–P–406 shall be employed for the water-absorption test to determine conformance with 3.6.6. Three pieces, 1 by 3 inches, shall be cut from the shells of the hats prepared in accordance with 4.6.1 and after subjection to one of the tests specified in 4.6.3, 4.6.4, or 4.6.5. Water absorption reported shall be the average of that determined for the three specimens.

4.6.8 *Headband clearance measurement.* To determine conformance with 3.4.2.2 clearance between the headband and helmet shell shall be determined by measuring the circumference of the outside of the headband (at maximum size for adjustable headbands) and of the inside of the shell at the juncture of the dome and the visor and computing the diameters and clearance by the following formulas:

$$\text{Diameter} = \frac{\text{Circumference}}{\pi}$$

$$\text{Clearance} = \frac{D-d}{2}$$

D—inside diameter of shell

d—outside diameter of headband (maximum size)

4.6.9 *Weight.* Helmets shall be weighed complete without lining and chinstrap on any suitable weighing balance to the nearest tenth (.1) ounce.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

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5.1 Packaging.

5.1.1 *Level A.* Each helmet shall be wrapped in paper conforming to UU–P–268 (grade B, basis weight 30 pounds). The paper shall be molded over the crown of the helmet, extend around the brim, and be tucked on the inside of the crown. Five wrapped helmets shall be stacked together by nesting the crowns one within the other.

5.1.2 *Level C.* Helmets shall be packaged according to the supplier's commercial practice.

5.2 Packing.

5.2.1 Each shipping container (level A or B) shall be provided with a full-height slotted partition made of double-faced corrugated fiberboard which shall divide the box into four equal sections. The container and inner packing shall be designed so that a stock of five packaged helmets shall fit snugly in the partitioned section of the box.

5.2.2 *Level A.* Helmets, packaged as specified in 5.1.1 and 5.2.1, shall be packed in a box conforming to PPP–B–591 (overseas type, style B), PPP–B–621 (class 2, style 4 for type 2 load), PPP–B–585 (style 1, class 3 for type 2 load), MIL–B–10377 (overseas type, style B), or PPP–B–601 (overseas type, style B using type III, class 1 plywood). Boxes shall have case liners conforming to MIL–L–10547 (type I, grade B, class 2), closed and sealed in accordance with the appendix thereto. Boxes shall be closed and strapped in accordance with the appendix to the applicable specification. Or, helmets packaged as specified in 5.1.1 and 5.2.1, shall be packed in a box conforming to PPP–B–636 (type I, class 2, grade 3, style RSC for type 2 load). All joints and corners of boxes shall be sealed with tape conforming to PPP–T–76 (2-inch). Boxes shall be banded with tape conforming to PPP–T–97 (type III), applied in accordance with the appendix thereto.

5.2.3 *Level B.* Helmets packaged as specified in 5.1 and 5.2.1 shall be packed in a box conforming to PPP–B–591 (domestic type, style B), PPP–B–621 (class 1, style 4 for type 2 load), PPP–B–585 (style 2, class 1 for type 2 load), PPP–B–601 (domestic type, style B using type III, class 1 plywood), MIL–B–10377 (domestic type, style B), or PPP–B–636 (type I, class 1, style RSC, for type 2 load). Boxes shall be closed in accordance with the appendix to the applicable specification.

5.2.4 *Level C.* Helmets packaged as specified in 5.1.2 shall be packed according to the contractor's commercial domestic practice to assure acceptance by carrier and safe delivery to destination, at the lowest transportation rate or rating. Containers shall comply with carrier rules and regulations applicable to the mode of transportation without assessment of penalty charges for improper packing.

5.2.5 *Level D.* Helmets packaged as specified in 5.1.2 shall be packed according to the contractor's commercial export practice to assure acceptance by carrier and safe delivery to destination at the lowest transportation rate or rating. Containers shall comply with carrier rules and regulations applicable to the mode of transportation without assessment of penalty charges for improper packing.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to marking required in the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (see 1.2.1).

(*c*) If adjustable chinstrap and winter lining are required (see 3.2 and 3.4.4).

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(*d*) If a specific shell material is required (see 3.3.1).

(*e*) If artificial leather headband is desired (see 3.3.2).

(*f*) Color and color number, when required (see 3.5).

(*g*) Quantity of each required.

(*h*) Applicable levels of packaging and packing required (see 5.1 and 5.2).

6.1.1 When headband assemblies (for replacement purposes) are ordered, the manufacturer's type or model number of the helmet for which the replacement headband is ordered should be stated, or if possible a representative helmet should be furnished in order that the manufacturer may have the necessary identifying information.

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL GGG–H–142, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.4 Standard head forms. The standard head forms mentioned throughout this specification are the model heads known as "A.M.L. Head Size Standards", a set of which is located at each of the following places:

115 Industrial Building
National Bureau of Standards
Washington, D.C.

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio

Material Laboratory
New York Naval Shipyard
Brooklyn 1, New York

6.5 For convenience, table I shows the transmitted forces for the measured impression diameters on impression bars ranging in Brinell hardness from 18 to 30 as computed from the Brinell formula (see 4.6.3.5).

6.6 Transportation description. Transportation descriptions and minimum weights applicable to this item are:

Helmets, Safety, Other Than Military, NOIBN.—Carload minimum weight 12,000 pounds, subject to Rule 34, Uniform Freight Classification.

Helmets, Welders'.—Motor volume minimum weight 10,000 pounds.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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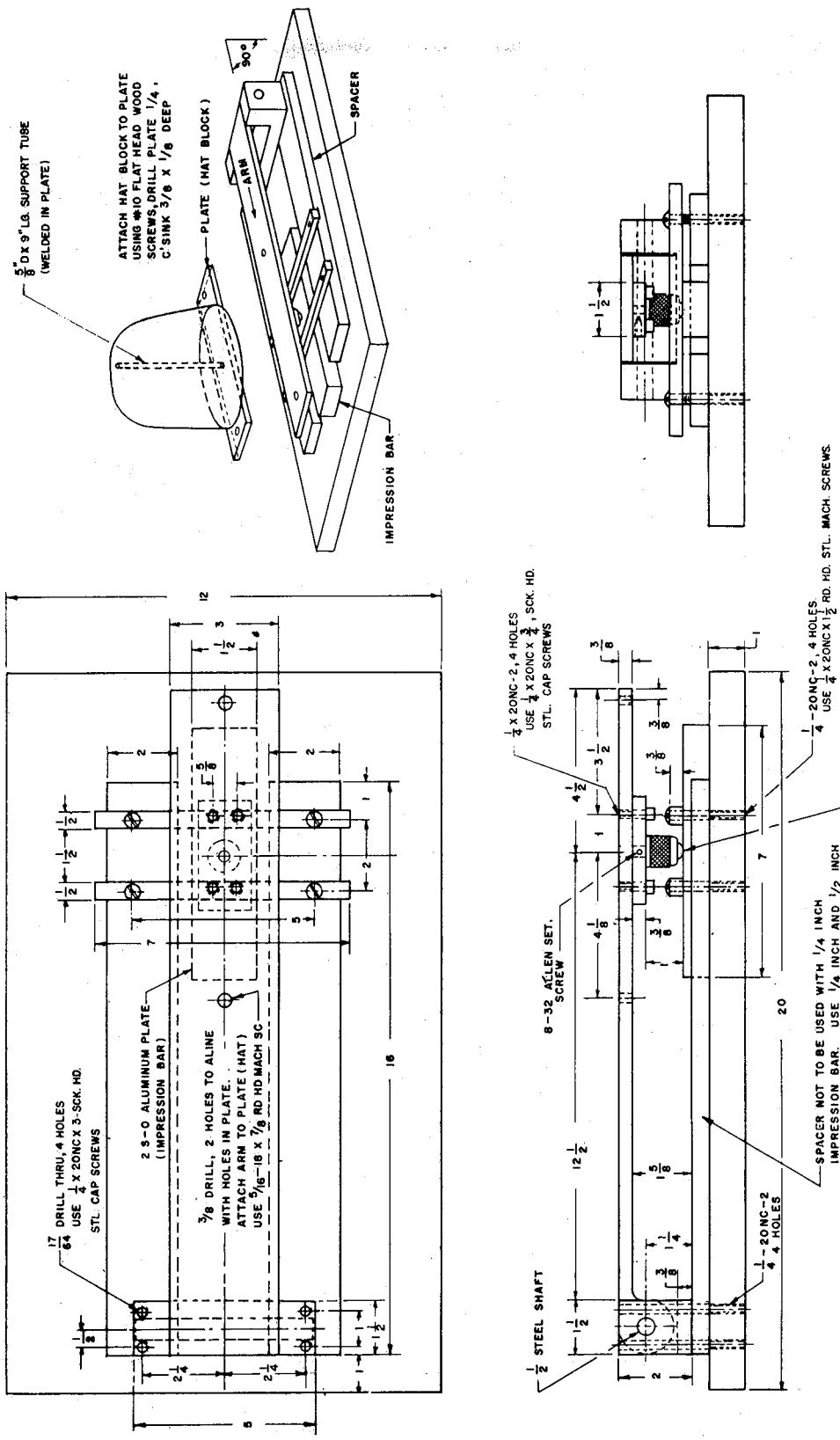


FIGURE 1.—Brinell hardness penetrator assembly

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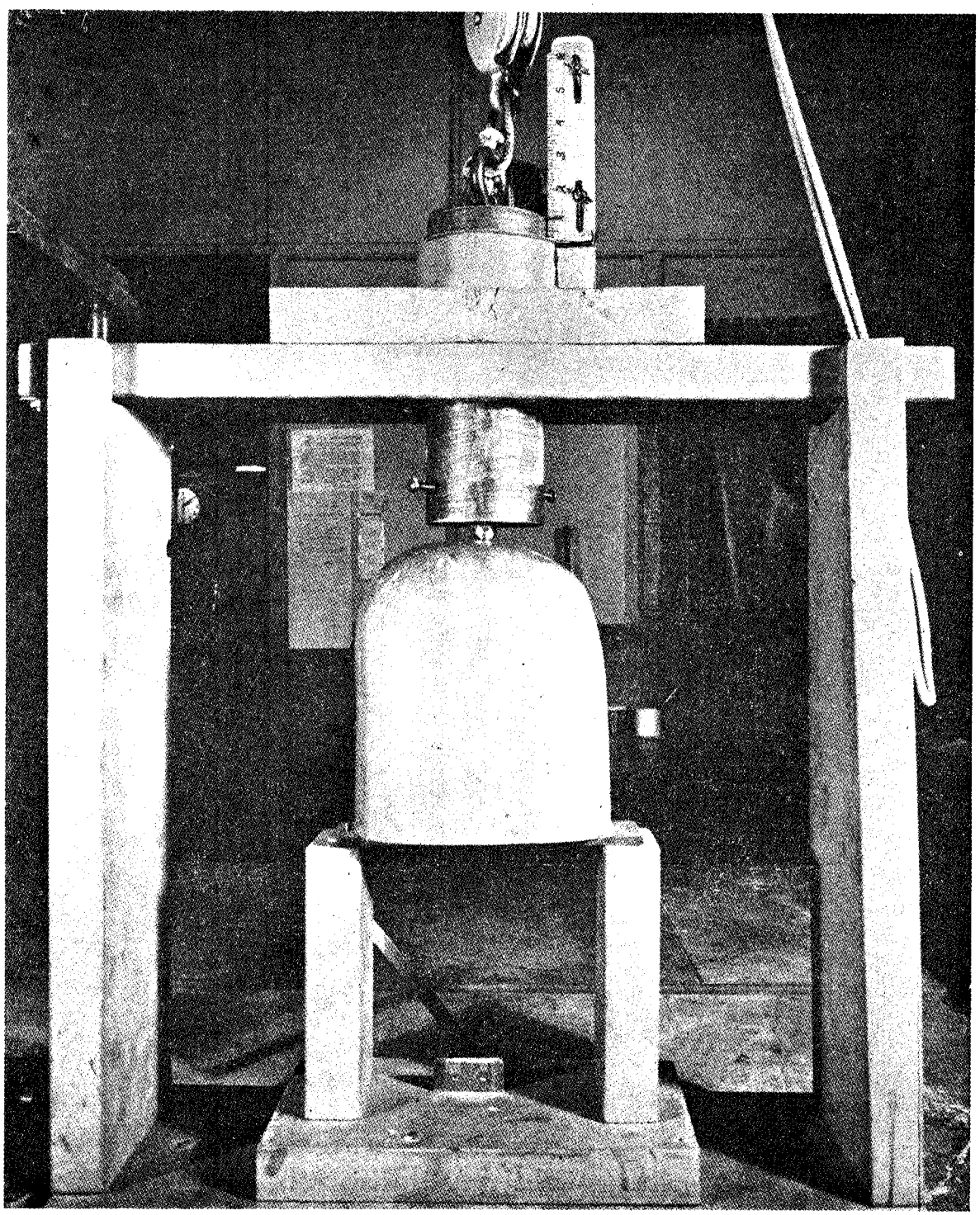


FIGURE 2.—Brinell hardness penetrator

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TABLE I.—*Transmitted forces in pounds.*

Diameter of impression, millimeters	Brinell hardness number												
	18	19	20	21	22	23	24	25	26	27	28	29	30
3. 9	485	515	540	565	595	620	650	675	700	730	755	785	810
4. 0	510	540	570	600	625	655	680	710	740	765	795	825	850 A
4. 1	540	570	600	630	660	690	720	750	775	810	835	865	895
4. 2	565	600	630	660	690	720	755	785	815	850	880	910	940
4. 3	595	625	660	695	725	760	790	825	860	890	925	955	990 B
4. 4	625	660	690	725	760	795	830	865	900	935	970	1000	1040
4. 5	650	690	725	760	795	835	870	905	940	975	1015	1050	1085
4. 6	680	720	760	795	835	870	910	950	985	1025	1060	1100	1135
4. 7	715	755	790	830	870	910	950	990	1030	1070	1110	1150	1190
4. 8	745	785	825	870	910	950	990	1035	1075	1115	1155	1200	1240
4. 9	780	820	865	905	950	995	1035	1080	1125	1165	1210	1255	1295
5. 0	810	850	900	945	990	1040	1080	1130	1175	1220	1265	1310	1355
5. 1	A 845	895	940	985	1035	1080	1130	1175	1220	1270	1315	1365	1410
5. 2	880	930	980	1030	1080	1125	1175	1225	1275	1325	1370	1420	1470
5. 3	920	970	1020	1070	1120	1175	1225	1275	1325	1375	1430	1480	1530
5. 4	955	1005	1060	1115	1165	1220	1270	1325	1380	1430	1485	1535	1590
5. 5	B 990	1045	1100	1155	1210	1265	1320	1375	1430	1485	1540	1595	1650
5. 6	1030	1085	1140	1200	1255	1310	1370	1425	1480	1540	1600	1655	1710

Notes: Values below line A are above or equal to the specified average force.
Values below line B are above the specified individual force.

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